

LAB 2 - MONOLITHIC ARCHITECTURE

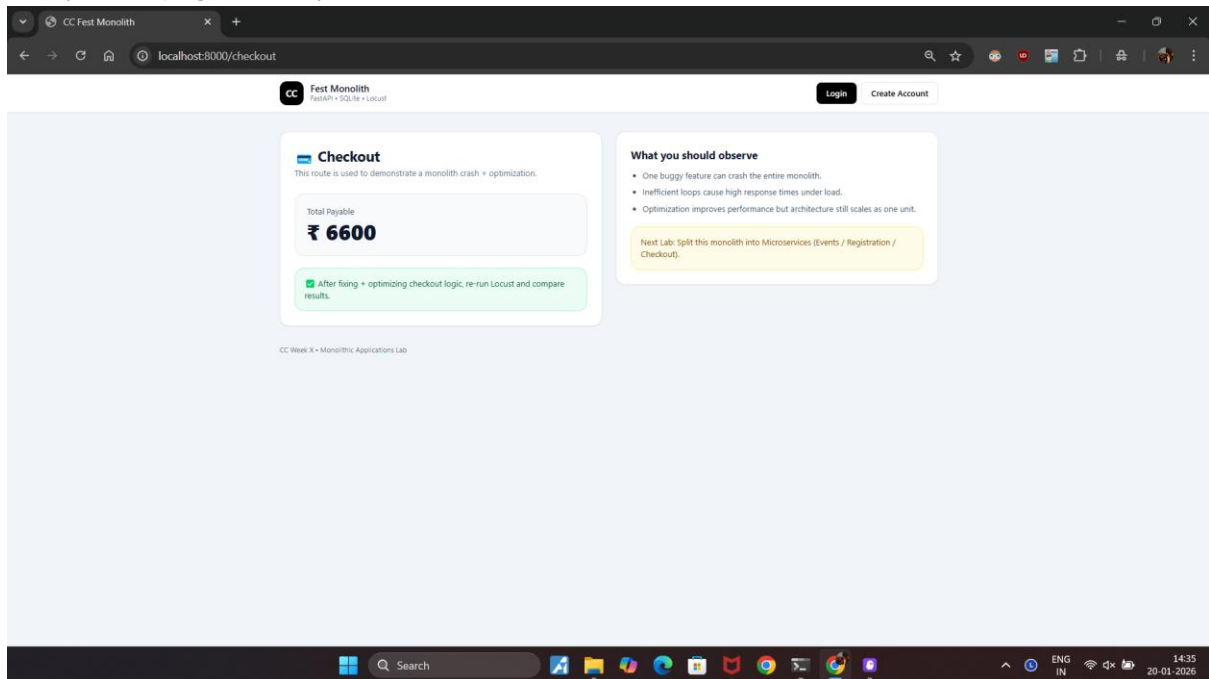
Name : Deepesh Padhy

SRN : PES2UG23CS163

SEC : 'c'

Github repo link : https://github.com/deepeshh7/CloudComputing_Lab2

SS1 (Events page loaded)



SS2 – monolithic failure

localhost:8000/events?user=PES2UG23CS163

CC Fest Monolith
PES2UG23CS163 + Logout

Logged in as PES2UG23CS163 Events My Events Checkout Logout

Events

Welcome PES2UG23CS163. Register for events below.

[View My Events](#)

Event ID: 1 ₹ 500

Hackathon

Includes certificate • instant registration • limited seats

[Register](#)

Event ID: 2 ₹ 300

Dance

Includes certificate • instant registration • limited seats

[Register](#)

Event ID: 3 ₹ 500

Hackathon

Includes certificate • instant registration • limited seats

[Register](#)

Event ID: 4 ₹ 300

Dance Battle

Includes certificate • instant registration • limited seats

[Register](#)

Event ID: 5 ₹ 400

AI Workshop

Includes certificate • instant registration • limited seats

[Register](#)

Event ID: 6 ₹ 200

Photography Walk

Includes certificate • instant registration • limited seats

[Register](#)

Event ID: 7 ₹ 350

Gaming Tournament

Includes certificate • instant registration • limited seats

[Register](#)

Event ID: 8 ₹ 250

Music Night

Includes certificate • instant registration • limited seats

[Register](#)

Event ID: 9 ₹ 150

Treasure Hunt

Includes certificate • instant registration • limited seats

[Register](#)

Event ID: 10 ₹ 300

Stand-up Comedy

[Register](#)

Event ID: 11 ₹ 450

Robo Race

[Register](#)

Event ID: 12 ₹ 500

Hackathon

[Register](#)

localhost:8000/register_event/2?user=PES2UG23CS163

```
INFO: 127.0.0.1:61976 - "GET /checkout HTTP/1.1" 500 Internal Server Error
ERROR: Exception in ASGI application
```

SS3 -

localhost:8000/checkout

CC Fest Monolith
PES2UG23CS163 + Logout

[Login](#) [Create Account](#)

Monolith Failure HTTP 500

One bug in one module impacted the entire application.

Error Message
division by zero

Why did this happen?

Because this is a **monolithic application**: all modules share the same runtime and deployment. When one feature crashes, it affects the whole system.

What should you do in the lab?

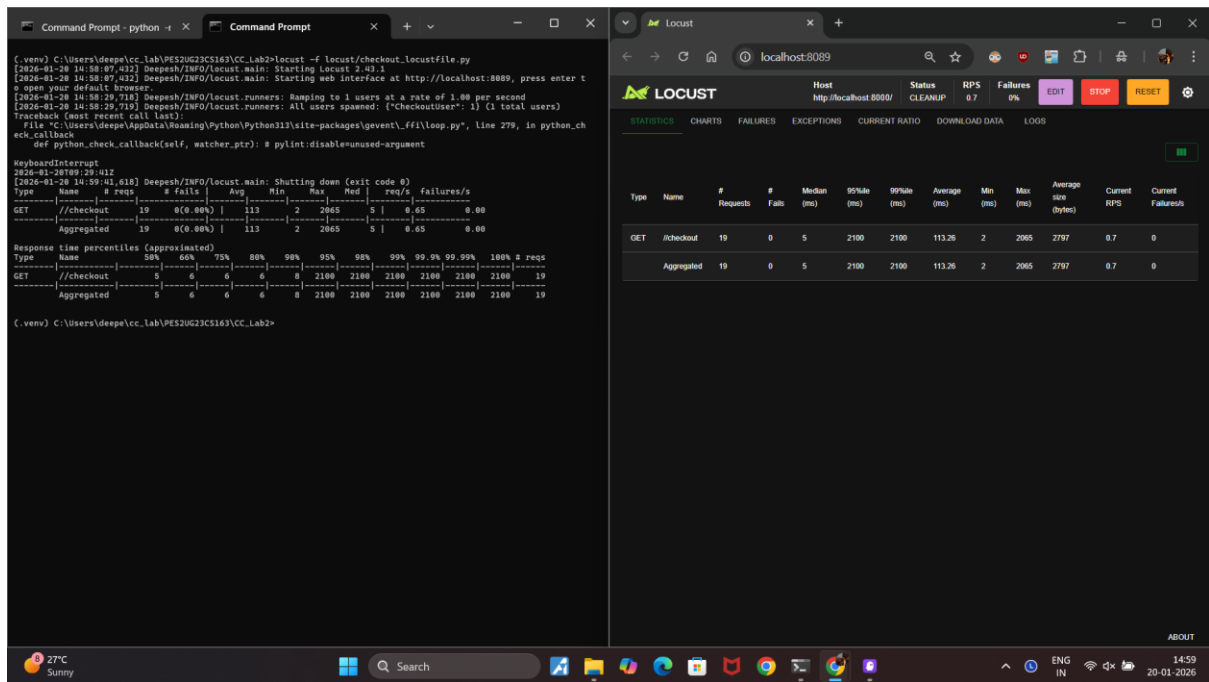
- Take a screenshot (crash demonstration)
- Fix the bug in the indicated module
- Restart the server and verify recovery

[Back to Events](#) [Login](#)

CC Week X • Monolithic Applications Lab

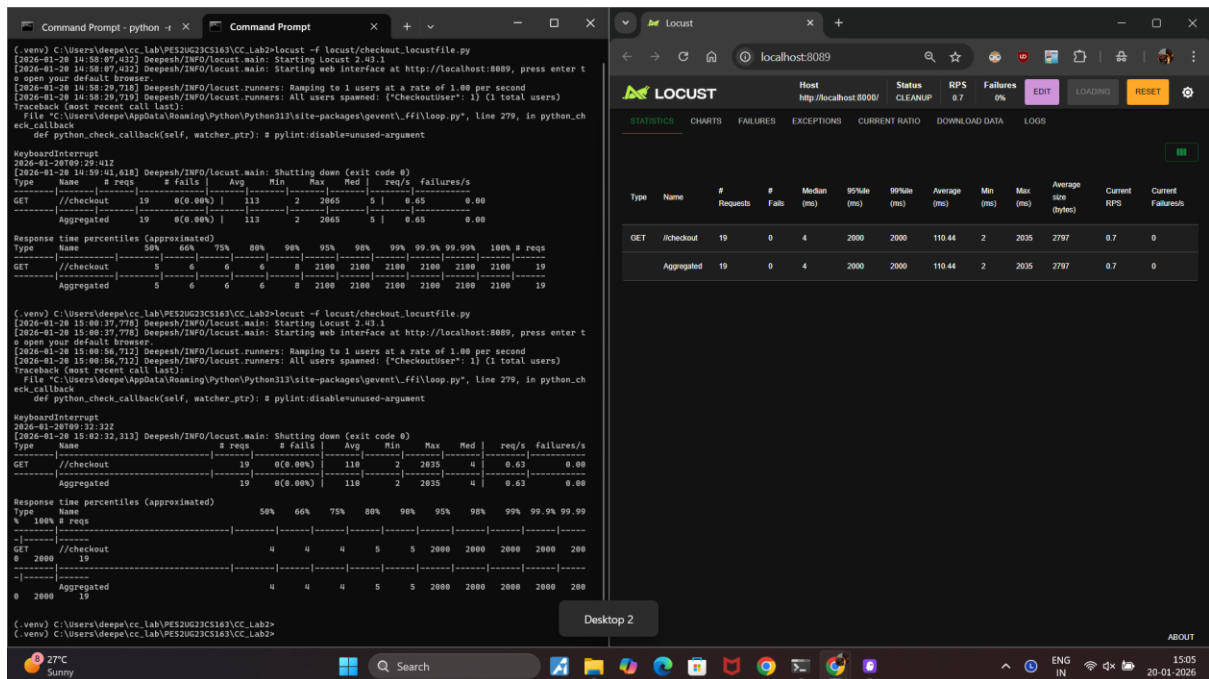
```
INFO: 127.0.0.1:61941 - "GET /checkout HTTP/1.1" 200 OK
```

SS4

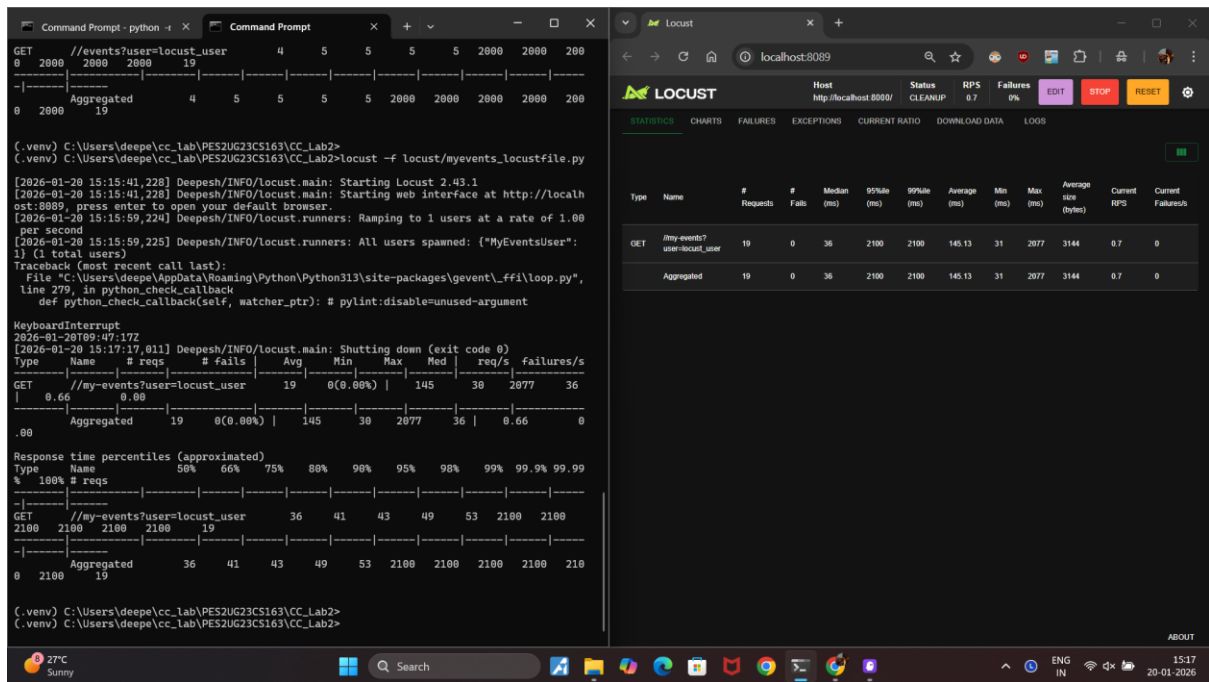


Route 1: /events

BEFORE optimization → SS6



AFTER optimization → SS7



Short explanation of optimizations

/events route

1. What was the bottleneck?
Inefficient looping and repeated processing of event data.
2. What change did you make?
Optimized the logic by reducing unnecessary loops and computations.
3. Why did the performance improve?
Less processing per request reduced response time.

/my-events route

1. What was the bottleneck?
Slow filtering and aggregation of user-specific events.
2. What change did you make?
Simplified the code to fetch and process only required data efficiently.
3. Why did the performance improve?
Fewer operations led to faster execution and lower latency.